

The empirical recurrence for OEIS A234401: recovering a conjecture that is not written down

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Abstract

OEIS A234401 records “Empirical recurrence of order 60”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 343$ vertices and satisfies some monic recurrence of order at most S ; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 60, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 60 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A234401 is “Number of $(n+1) \times (2+1)$ 0..6 arrays with every 2×2 subblock having the sum of the absolute values of the edge differences equal to 12.”. It has offset 1 and begins

5646, 204320, 6467216, 220295536, 7430382692, 252526817148, 8579502870748, 291761755509076, ...

The entry states:

Empirical recurrence of order 60 (see link above)

As of the “Last modified” line on the live entry (Jun 02 2025 09:17:35, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 343$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S = 343$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 686$ exact terms of the model returns order 60 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{60} c_i a(n-i),$$

c_1	42	c_{16}	3936825640499573	c_{31}	597751333083684366122	c_{46}	−33841835737
c_2	230	c_{17}	6958169582892303	c_{32}	9917735778187692021358	c_{47}	−9488316174
c_3	−21067	c_{18}	−62475879129069204	c_{33}	−1727886733729875623913	c_{48}	77033758409
c_4	41293	c_{19}	−71169403049524908	c_{34}	−19760864364857975090600	c_{49}	29873793261
c_5	4000778	c_{20}	749774019483732999	c_{35}	3176536726534157180880	c_{50}	−1050255765
c_6	−16833125	c_{21}	523289915169517776	c_{36}	30262590935435818718596	c_{51}	−5796871142
c_7	−397423643	c_{22}	−6893718414265243632	c_{37}	−3649383718048213895738	c_{52}	5998651224
c_8	2101076834	c_{23}	−2640256308926852298	c_{38}	−35384080686740227114748	c_{53}	6523611075
c_9	23864708139	c_{24}	48958265948927178924	c_{39}	2113259987338312204508	c_{54}	300293815
c_{10}	−142885331062	c_{25}	7765225278966119342	c_{40}	31290944166567450854893	c_{55}	−36454878
c_{11}	−940143766205	c_{26}	−269781713576705705527	c_{41}	485084961544187217346	c_{56}	−55170081
c_{12}	618382224651	c_{27}	−737725396313766912	c_{42}	−20639330047693813210244	c_{57}	57676059
c_{13}	25565496208261	c_{28}	1155480337277717809552	c_{43}	−2047377089273108836864	c_{58}	17749677
c_{14}	−183925178822995	c_{29}	−114143831266724081789	c_{44}	9942022999398870922577	c_{59}	10361732
c_{15}	−495330773750106	c_{30}	−3844835914979772049890	c_{45}	1863208165574324170492	c_{60}	262580

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 60, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $60 + 4$ terms removed returns order 60 as well, so $\deg q_{\min} = 60 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 13 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $60 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 60 that this sequence satisfies — and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A234401.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.